

Investigating the Minimal Condition of the Dynamic Invisible Body Illusion

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ABSTRACT

Visual-tactile synchronization or visual-motor synchronization makes a non-innate body feel like one's own body (illusory body ownership). A recent study has shown that body ownership is induced to an empty space between the hands and feet from the motion of only hands and feet (dynamic invisible body illusion). However, it is unclear whether both hands and feet are necessary to induce the illusion. In this study, we investigated the minimal condition for the dynamic invisible body illusion by manipulating the presentation of hands and feet. Our results suggest that both hands and feet are necessary for the illusion.

Keywords: Body ownership, self-location, invisible body

Index Terms: I.3.7 [Computer graphics]: Three-Dimensional Graphics and Realism—Virtual Reality; H.1.2 [Models and Principles]: User/Machine Systems—human factors

1 INTRODUCTION

Visual-tactile synchronization [1] or visual-motor synchronization [2] makes a non-innate body feel like one's own body. For example, when the back of a virtual body and a participant's back are simultaneously stroked, and the participant observes the back of the virtual body through a head-mounted display (HMD), the participant feels as if it was their own body [1]. In addition, when body ownership is generated in the front body, the participant's self-location drifts forward [1].

Interestingly, body ownership is also induced in an invisible full-body. For example, when a participant's body and empty space are stroked with a brush simultaneously, the sense of ownership of an invisible full-body is elicited [3]. Also, Kondo et al. [4] showed that presenting only hands and feet moving in synchronously with a participant's voluntary movements made an empty space between the hands and feet feel like the participant's own body (dynamic invisible body illusion).

However, it is not clear whether both hands and feet are necessary to induce the dynamic invisible body illusion. In this study, we investigated the minimal condition to induce the dynamic invisible body illusion by manipulating the presentation of hands and feet in a virtual environment. In the experiment, we used a questionnaire to measure body ownership and measured drifts in self-location of shoulders and knees. For example, if body ownership of an invisible body was generated by hands alone, we expected that drifts in self-location would occur not only on

shoulders close to the hands but also on knees far from the hand. We hypothesized the same for feet-only stimulation.

2 METHODS

Twenty-four volunteers participated in the experiment. The sample size was calculated using G*Power 3.1 (repeated measures ANOVA, three avatar conditions, medium effect size $f = 0.25$, $\alpha = 0.05$, power $(1 - \beta) = 0.8$). They had a healthy vision and motor abilities. The study was conducted with the approval of the Ethical Committee on Human-subject studies of Keio University.

Participants wore motion capture suits, and their movements were tracked by 24 motion capture cameras (OptiTrack). Participants wore an HMD (HTC Vive Pro Eye, 1440 x 1600 pixel, 90Hz, 110deg) to observe a computer-generated virtual environment and held two Vive controllers. The participants observed gloves and socks that moved synchronously with the participant's movement (GSN: Gloves Socks Normal, Figure 1 left), gloves and socks that moved synchronously but were shifted 25 cm to the left (GSS: Gloves Socks Shift), gloves only that were shifted 25 cm to the left (Gloves), or socks only that were shifted 25 cm to the left (Socks) from a first-person perspective in the virtual environment. A mirror in front of the participant in the virtual environment reflected the gloves and socks. In order to measure drifts in the self-location, the gloves and socks were shifted to the left. It has been reported that a shift of 25 cm did not weaken body ownership[5].

The mental imagery task (MIT) [1] was used in the experiment to measure self-location. MIT used a modified version of the method of the previous study [1], with balls coming from the left and right. In MIT, a ball rolled toward the participant and the screen went dark 3 s after the ball began to move. After the screen went dark, the participants continued to predict the ball's movement and pressed a button when they felt it hit their body.

In our experiment, the participants practiced MIT for 12 trials (two directions (left, right) x two targets (shoulder, knee) x three repetitions). Then they completed MIT for 12 trials as a baseline. Next, the participants moved the gloves and socks according to the sound for 2 min to induce body ownership (learning session). After that, they performed the MIT again for 12 trials. Finally, participants answered a questionnaire on a seven-



Figure 1: (Left) Virtual gloves and socks. (Right) Results of MIT. Error bars indicate SE.

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level Likert scale ranging from -3 (strongly disagree) to +3 (strongly agree). Each item in the questionnaire was presented in random order. The procedure from the learning session to the questionnaire was conducted four times, once for each body condition (four avatar conditions x one repetition), in random order.

3 RESULTS

3.1 MIT

The results of MIT for each condition, minus the baseline results, were analyzed as the drift of self-location. The results showed no significant differences between any of the conditions (Figure 1 right).

3.2 Questionnaire

The gloves and socks in the same position as the participant induced stronger body ownership of the space between the gloves and socks than left-shifted gloves only and left-shifted socks only (Figure 2: Q1 GSN vs. Gloves: $z = 3.90$, $p < .001$, $r = .80$; GSN vs. Socks: $z = 3.56$, $p < .001$, $r = .73$). The left-shifted gloves-socks also showed stronger body ownership than the left-shifted gloves-only (Q1 GSS vs. Gloves: $z = 3.03$, $p = .010$, $r = .62$). There was no significant difference in body ownership between GSN and GSS conditions (Q1 GSN vs. GSS: $z = 2.45$, $p = .075$, $r = .50$). Q2 reported a higher agency in all conditions, with no significant differences. Q3 showed a similar trend to Q1; the participants perceived a full-body in the empty space more strongly in GSN condition than Gloves condition and Socks condition (Q3 GSN vs. Gloves: $z = 3.44$, $p = .001$, $r = .70$; GSN vs. Socks: $z = 3.34$, $p = .002$, $r = .68$). The participants also perceived a full-body in the empty space more strongly in GSS condition than Gloves condition (Q3 GSS vs. Gloves: $z = 2.65$, $p = .044$, $r = .54$). Q4 and Q5 were control questions, with no significant differences between conditions.

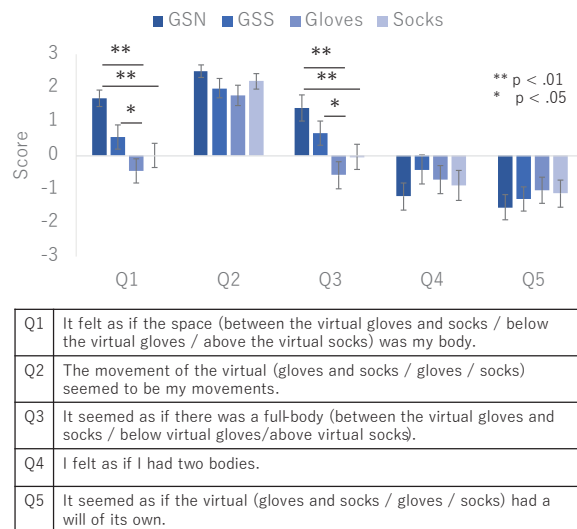


Figure 2: Results of questionnaire. Error bars indicate SE. Words in brackets were used according to avatar conditions.

4 DISCUSSION

Our study investigated the minimal condition for the dynamic invisible body illusion by manipulating the presentation of the gloves and socks. The results showed that the gloves or socks alone elicited lower body ownership of the invisible body than the

condition with both gloves and socks. This suggests that both gloves and socks are necessary to induce the dynamic invisible body illusion.

However, since the body ownership was strongest in the condition that the gloves and socks were not shifted, the shift of the gloves and socks may have attenuated the body ownership. In the present study, gloves and socks were shifted for the drift measurement based on research [5] that shows that shifting at 25 cm does not decrease body ownership. However, body ownership in the shifted gloves and socks condition was weak, and it was not significantly different from the shifted sock-only condition with no body ownership (there was a significant difference between unshifted gloves-socks and shifted socks-only). Thus, even a 25-cm shift of the body parts would seem to attenuate the body ownership in the invisible body.

In Q3, regarding the perception of the invisible body, the participants reported that they perceived the full-body more strongly when both gloves and socks were present. This is a similar trend to the question on body ownership (Q1), and the perception of the full-body in an empty space might be important in the process to induce the ownership of invisible body. The same trend has been reported in the previous dynamic invisible body illusion study [4].

Our study used MIT [1] to measure drifts in self-location, but no significant difference existed between the conditions. The paper conducted with MIT [1] used a full-body avatar, so the drift may not be caused in invisible body parts. Another possible reason that the body ownership score for the shifted gloves and socks in this study was weak, at approximately 0.5.

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